Akarsh Pandey

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Electronics and Communication Engineering
Bundelkhand Institute of Engineering and Technology , Jhansi

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Education

- **Bundelkhand Institute of Engineering and Technology** — *2024-present*
  B.Tech. in Electronics and Communication Engineering — CGPA: 7.72 (till 4th semester)
- **The Aryan International School** — *2023*
  Intermediate, Central Board of Secondary Education, UP — Percentage: 85
- **International Hindu School** — *2021*
  High School, Central Board of Secondary Education, UP — Percentage: 91

Work Experience

- **Project Intern | C3iHub, IIT Kanpur** **Link** — *June 2026 – Nov 2026*
  Project: Automated Firmware Analysis on DLMS / COSEM Protocol
  - Worked on the research and development of an automated firmware analysis framework for DLMS/COSEM-compliant smart meters used in Advanced Metering Infrastructure (AMI).
  - Studied firmware architecture and security mechanisms to identify potential vulnerabilities in smart meter firmware before deployment.
  - Explored static and dynamic firmware analysis techniques, including reverse engineering, symbolic execution, and fuzz testing for vulnerability assessment.
  - Developed a proof-of-concept automated firmware analysis pipeline for DLMS/COSEM firmware, integrating reverse engineering and AI-assisted vulnerability detection techniques.

Projects

- **AI-Based Hand Gesture Controlled Robotic Car** **Link** — *2025*
  Built a Computer Vision-based robotic car using real-time hand gesture recognition.
  - Tools & technologies used: Python, OpenCV, MediaPipe, Arduino UNO, Serial Communication, L298N Motor Driver, HC-05 Bluetooth Module
  - Developed a real-time hand gesture recognition system using OpenCV and MediaPipe to detect finger gestures from a laptop camera.
  - Mapped hand gestures to robotic car movements (Forward, Backward, Left, Right, Stop) by communicating commands from Python to Arduino UNO via serial communication.
- **Linear Regression vs Random Forest Comparison on World Happiness Dataset** **Link** — *2026*
  Compared machine learning regression models for predicting happiness scores using real-world data.
  - Tools & technologies used: Python, Pandas, NumPy, Matplotlib, Scikit-learn, Jupyter Notebook
  - Performed data preprocessing, feature engineering, exploratory data analysis (EDA), and trained Linear Regression and Random Forest models.
  - Evaluated model performance using MAE, MSE, RMSE, and R^2 score, demonstrating superior predictive accuracy of Random Forest over Linear Regression.

Technical Skills and Interests

Languages: Python , C
Developer Tools: Jupyter Notebook, VS Code
Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn
Cloud/Databases: MySQL, CSV/Excel Data Handling
Soft Skills: Problem Solving, Quick Learner, Analytical Thinking, Team Collaboration, Communication
Coursework: Data Analytics, Machine Learning Basics, Digital Electronics
Areas of Interest: Machine Learning, Data Analytics, AI-based Projects, Electronics and Embedded Systems

Positions of Responsibility

- **Member,** Table Tennis Sports Sub Council — *October 2024 - present*
- **Coordinator,** Innovation and Incubation Centre — *October 2025 - present*